

# Christophe de Dinechin

Software as a Passion

LANGUAGES | SYSTEMS | GAMES | PHYSICS | ART | TOOLS

## ◈ ABOUT

Pushing the boundaries by writing highly innovative and differentiating software to make as much of a difference as possible.

French software developer — games, languages, operating systems

Amateur physics researcher

Tinkerer in strange forms of art

## ◎ KEY SKILLS

- ✓ Languages & Compilers
- ✓ Systems & Virtualization
- ✓ Games & 3D
- ✓ Physics
- ✓ Art & Culture
- ✓ Tools & Apps

## ▤ TECHNOLOGIES

C++ · C · RPL · Lisp · XL · Tao3D · ELFE · Linux · macOS · Windows · SPICE · Emacs · OpenShift · Make

## ℹ INFO


- ★ Tinkerer in strange forms of art
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**Scan the constellation**
https://c3d.github.io/cv/

## LANGUAGES & COMPILERS

### HP aC++ Compiler & C++ ABI — 1998-2000

- Worked on C++ templates and exception handling
- Initiated / contributed to the common C++ ABI for Itanium

### DB48X - Modern RPL Calculator — 2022-Present

- Rebuild and improve the HP48-family user experience
- Rich RPL data types and built-in functions with Lisp-level programming power

## SYSTEMS & VIRTUALIZATION

### DxO ONE Connected Camera — 2015-2016

- Dual-core ARM + ISP real-time system
- Improving software quality on a highly capable connected camera

### HPVM - Enterprise Virtualization — 2000-2010

- Started the HP Integrity Virtual Machines project in late 2000
- Early bring-up with Todd Kjos and Jonathan Ross (4-way HP-UX guests)

## GAMES & 3D

### Alpha Waves / Continuum — 1989-1990

- Started in 1988, published by Infogrames in 1990
- First 3D platform game; true 6-axis 3D with physics-like interaction

### HP-48 Games & HPDS — 1989-1993

- HPDS — HP Development System for HP calculators (from earlier SATAS)
- PacMan — first use of HP48 hardware scroll (per Paul Courbis)

## PHYSICS

### Physics Research & Books

- A Theory of Incomplete Measurements — unifying GR and quantum mechanics
- New relativity principle: do physics with any instrument

## ART & CULTURE

### 3DIVART

- Adapt ray-tracing / ray-marching shaders for auto-stereoscopic screens via Tao3D
- Synthetic 3D art you can see without glasses on suitable displays

### Informagie

- Started as a student around the same time as Alpha Waves
- Finished after 32 years and published

## TOOLS & APPS

### Make-It-Quick

- Short readable makefiles without automake/cmake complexity
- Automatic incremental configuration (`config.h`)

### Recorder

- Continuous recording with printf-like statements
- Classify events to keep both old important and recent rapid events